

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 3_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 15

Section 1 : MCQ

1. The main idea behind the Knuth-Morris-Pratt algorithm is to:

Answer

Avoid redundant checking of characters that have already been matched

Status : Correct

Marks : 1/1

2. The Boyer-Moore algorithm uses which two rules to determine the shift amount?

Answer

Good suffix rule and bad character rule

Status : Correct

Marks : 1/1

3. The Boyer-Moore algorithm is particularly efficient when

Answer

The pattern contains repetitive elements

Status : Correct

Marks : 1/1

4. In the KMP algorithm, if the prefix table at position i has a value of k , what does it imply?

Answer

The length of the longest proper prefix which is also a suffix up to position i is k .

Status : Correct

Marks : 1/1

5. The concept of prefix and suffix is used in which of the following algorithms?

Answer

Knuth-Morris-Pratt (KMP) algorithm

Status : Correct

Marks : 1/1

6. What is the main purpose of Manacher's Algorithm?

Answer

To find the longest palindromic substring in linear time

Status : Correct

Marks : 1/1

7. In the Z-algorithm, the Z-array is used to store:

Answer

The length of the longest substring starting at each position that matches the prefix

Status : Correct

Marks : 1/1

8. Which application is most efficiently solved using Z-algorithm?

Answer

Finding all occurrences of a pattern in a text

Status : Correct

Marks : 1/1

9. In Z-algorithm, when $Z[k]$ is greater than 0, what does it imply?

Answer

Substring starting at k matches the prefix for length $Z[k]$

Status : Correct

Marks : 1/1

10. Which of the following best explains why Manacher's algorithm is preferable for palindrome problems over expand-around-center approaches?

Answer

Avoids recomputation of overlapping palindromes by mirroring results around the center

Status : Correct

Marks : 1/1

11. In bioinformatics applications, KMP is often used instead of Boyer-Moore. Why?

Answer

DNA has only 4 characters, reducing Boyer-Moore's skipping advantage

Status : Correct

Marks : 1/1

12. In KMP, the LPS array for pattern "ABABAC" is:

Answer

[0,0,1,2,3,0]

Status : Correct

Marks : 1/1

13. Which is true about Manacher's algorithm with respect to handling even-length palindromes?

Answer

Uses a modified string with separators between characters

Status : Correct

Marks : 1/1

14. How is Z-algorithm used for pattern search in text efficiently?

Answer

Concatenate P + "\$" + T and compute Z-array

Status : Correct

Marks : 1/1

15. Which of the following strings has the largest number of palindromic substrings using Manacher's algorithm?

Answer

"aaaaa"

Status : Correct

Marks : 1/1